



Farmers are advised to follow the sowing rule to take-up sowing of rainfed crops like Cotton, Soybean, Maize, Jowar, Redgram, Greengram etc., only after receipt of cumulative rainfall of 50-60 mm in light soils and 60-75 mm in heavy soils or the soil should be wet up to top 15-20 cm depth after enter of South-West monsoon rains into the state.

CEREAL CROPS

Paddy

Under assured irrigation farmers are advised to raise rice nurseries of long duration (140-145 days) varieties from May 20th to June 20th. Under assured irrigation farmers are advised to raise rice nurseries of medium duration (130-135 days) or short duration (120-125 days) varieties from June 20th onwards. Month of June is not optimum for raising paddy nurseries using Telangana Sona (RNR15048) variety. Take the Greengram as catch crop to preceding rice in areas where the release of water into the canals and tanks is delayed. Green manuring with Sesbania/ Crotalaria / Pillipesara / grain legumes, residues of blackgram / greengram and in situ incorporation before flowering. Varieties: cultivation of long duration (BPT-5204, WGL-44, RNR 11718 etc.), medium duration (WGL-14, JGL-11470, JGL-384, RNR-28361, JGL-33124 etc), short duration (RNR-15048, KNM-1638, KPS-6251, WGL-362, KNM-118, JGL-24423, RNR-29325, JGL-28639 etc.) varieties.

Direct seeding of rice with Seed-cum ferti drill wherever suitable. In Telangana state, dry converted wet and wet direct seeded rice are popular, and the sowing can be started from June ending. Treat seeds with carbendazim @ 3 g/ kg (dry seed treatment) or 1 g/L (wet seed treatment). Farmers are advised to apply the Carbofuran 3CG granules @ 800 g per acre of nursery one week before pulling of the nursery. Root dipping: prepare the slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000 g/ha) of phosphobacteria or 10 packets of (2000 g/ha) of Azophos, inoculant in 40 lit. of water and dip. the root portion of the seedlings for 15-

30 minutes in bacterial suspension and transplant. Make 20 cm alleyways at every 2 meters to facilitate free aeration, ease of farm operations and better pest and disease management. Broadcast butachlor @ 1-1.5 l or pretilachlor @ 500-600 ml or 4 kg granules of bensulfuron methyl (0.6%) + pretilachlor (6.0%) with 20 kg sand per acre. Practice alternate wetting and drying in well irrigated areas to save water, better aeration and minimize incidence of BPH besides getting carbon credits with less emission of greenhouse gases. Recommended dose of fertilizers: 40-48 Kg Nitrogen, 20 Kg Phosphorus 16 Kg Potash per acre during Vanakalam. Spray 0.2% ZnSO₄ 2 to 3 times at 5 days interval to manage Zn deficiency, particularly in alkaline soils.

Maize

Deep summer plough to destroy pests and conserve moisture. Optimum sowing window is from June 15th to July 15th with medium and long duration varieties (90-100 days and 100-120 days) without any yield reduction. Sow medium duration varieties/hybrids DHM-206, DHM-182, DHM-117, Bisco-855, Bio-9637, Bisco-740, Kaveri Ekka-2288, Kaveri -2020, Kaveri-3696 and short duration hybrids DHM-115, Pioneer-3342, KH-5991, DKC-7074 during 15th June to 15th July. Sowing can be extended up to July end if monsoon is delayed. Sow on receipt of at-least 50-60 mm rainfall for better germination and complete sowing in a week for reducing the incidence of FAW. Spray NSKE 5% or neem oil 1500 ppm @5ml/l after sowing and at weekly intervals. Dibble seeds with tractor drawn cultivator or seed cum ferti drill on ridges and beds. Intercrop red gram varieties PRG-176, WRG-97 in 2:1 ratio in light soils and in 4:1 ratio in heavy soils to reduce incidence of FAW. Follow IPM for FAW management: summer ploughing, growing single cross hybrids, clean cultivation, intercropping with pulses, balanced fertilization, release of egg parasitoids (*Trichogramma* sp) and spraying Chlorantraniliprole @ 0.4 ml or spinetoram @ 0.5 ml or Emamectin benzoate @ 0.4 g per litre in plant whorls. As post emergence, spraying of Tembotrione @ 115 ml + Atrazine 400 g or Toperamezone @ 40 ml + Atrazine 400 g or mesotrione + atrazine @1400 ml/acre in 200 litre of water will effectively control weeds. After 30-35 days, crop may be inter-cultivated followed by earthing up is needed for better nutrient uptake and to avoid lodging.

COMMERCIAL CROPS

Cotton

Sow the crop early by June 1st/2nd week after receipt of sufficient good rains at least (60mm). Don't sow cotton crop after July 20th as yields will be reduced drastically under rainfed conditions. Sowing of Cotton on ridge and furrow in medium soil and broad Bed furrow in black soil under rainfed conditions facilitates the crop to moisture conservation and excess water drain out. In a village or area all the cotton cultivating farmers should take up the sowings on community basis and complete within a period of 7 - 10 days. Depending on soil type and rainfall pattern cotton should be sown at a spacing of 120 x 60 cm/ 120 x 45 cm/ 90 x 60 cm/ 90 x 45 cm/ 90 x 30 cm. Deep summer plough to destroy the eggs of insect pests and to remove weed stock. Intercrop with greengram/ blackgram/ soybean/ cluster bean in 1:2 or 1:3 ratio, and red gram 4:1 / 6:1 / 8:1 ratio which also helps to manage sucking pests.

Grow 2-3 rows of maize or sorghum as border crop as barrier to prevent pests. Treat the seed with imidacloprid 70 WS@ 5 g per kg or thiamethoxam 70% WS @ 4 g per kg followed by *Trichoderma viridi* or *Pseudomonas fluorescens* @ 10 g per kg.

Pre-emergence herbicides like pendimethalin (Stamp extra) @ 700 ml/acre should be applied within 24 -48 hours after sowing with the help of hand sprayer for better control of weeds in cotton. Application of post emergence herbicides like quizalofop ethyl @ 400 ml + pyriithiobac sodium @ 250 ml in 200 L of water at 30 DAS in cotton to control grasses and broad-leaved weeds. Inspect the crop at squaring and flowering stage for presence of PBW larvae within flowers. Installation of Pheromone traps @ 8/acre and monitoring ETL at this stage is 10 % damaged flowers. Spray $MgSO_4$ @ 10 g/l twice at 45 and 75 DAS; Boron @ 1.5 g/l twice at 60 and 90 DAS; 0.5% $FeSO_4$ with citric acid twice at weekly intervals and $ZnSO_4$ @ 2 g/l twice at 4-5 days interval at 45 DAS to manage micronutrient deficiencies. During excess rainfall, drain excess water and apply 25 kg urea and 20 kg potash per acre. Spray Planofix (NAA 10 PPM) @ 1 ml per 4.5 liters of water twice at 10-15 days interval to manage flower dropping. Spray 1% KNO_3 or 19:19:19, 20:20:20, 21:21:21. Follow rotation of insecticide sprays against insect pests for high efficiency. Stem application of fipronil 1: 4 at 30 DAS and Flonicamid / Imidacloprid with water (1:20) at 45, 60 DAS to be done which protects the crop from early season sucking pests like jassids and aphids.

PULSE CROPS

Pigeon pea

Sow high-yielding wilt-resistant varieties (155-165 days) like WRGE-93, WRGE-97, WRGE-121, and medium-duration varieties like WRG-255 and TDRG-4, TDRG-59 & TDRG-272 can be recommended. Sowing should be initiated only after receipt of 60 mm rainfall. The optimum sowing time should be 15th June to 15th July at a spacing of 150-180 x 20 cm in black soils and 90-120 x 20 cm in red soils. Seed rate should be 2.5 to 3.0 kg per acre only (plant-to-plant spacing should be mandatory). Treat seeds with thiram or captan @ 3g/kg followed by rhizobium culture @200-400 g/ 3 kgs of seed. The raised bed method or ridge and furrow method should be followed to protect the crop from *Phytophthora* blight disease, which occurs due to excess and stagnation of rainwater. Intercropping in Red gram + Cotton (1:4/ 1:6), Soybean (1:4/ 1:6), Green gram/ Black gram (1:7) should be followed. Treat the Red gram seed with *Trichoderma viride* @ 10 g/ kg seed to protect the crop from soil-borne diseases.

Spray pendimethalin @ 1 - 1.5 L in 200 liters of water within 24 - 48 hours after sowing will effectively control the weeds in Red gram Post-emergence application of quizalofop ethyl @ 400 ml/ acre to control grasses and imazethapyr 10 % @ 300 ml/ acre to control grasses and broad-leaved weeds. Both herbicides in the tank mix are also recommended in case both weeds are observed. If there is submergence due to excess rain, drain excess water at the earliest and spray 1% KNO_3 or 19:19:19, 20:20:20 or 21:21:21. Application of Multi K or 13-0-25 @ 10g/l at the time of flowering should be done. Practice nipping at 45 and 60 DAS, wherein terminal branches are removed to facilitate lateral branches, thus improving yields. To control pod fly spraying of Thiomethaxam 25 wg @ 0.2ml/l, Flubendamide 0.2ml/L with Jaggery 5g/L to be sprayed at the time of early pod formation stage.

Greengram and Blackgram

Sowing time is generally from June 15th fortnight- July 15th during Kharif, Sept 15th to 30th October during Rabi, January last week to March 15th during Summer. Greengram varieties MGG-295, MGG-385, MGG-351, TM-96-2, WGG-37 and WGG-42, Virat, IPM-2-14, Sikha. Blackgram YMV-resistant varieties MBG-1070, PU-31, GBG-1 and TBG-104 are the preferred varieties. Treat seeds with imidacloprid @ 5g or thiamethoxam @ 5g/kg. Intercropping can be done with cotton (2:1 or 1:1 ratio) or redgram (6:1)/(8:1). Greengram can be taken as a preceding crop to Rice and Chilli as green manure for incorporation in the soil.

OILSEED CROPS

Groundnut

Seed rate recommended is 237 kg/ha and practice seed treatment with imidacloprid 600 FS @ 2 ml/kg followed by tebuconazole 2 DS @ 1g or Trichoderma viride @ 10g/kg to control early and late leaf spot diseases, sucking pests, collar rot, PSND in the early stages. Application of Gypsum @ 500 kg/ha at pegging stage near podding zone for pod development. Weed management with Diclosulam 84% WDG @31 gm/ha as pre-emergence herbicide followed by post-emergence application of Imazethapyr @ 750 ml/ha at 20 to 25 days after sowing.

Soybean

Sow varieties ASB-22, JS-335, AISb-1, JS-93-05 during 15th June 1st week of July using seed drill/ gorru/ desi plough/ by tractor drawn seed cum fertilizer drill / broad bed furrow planter. Sowing of Soybean by Broad bed Furrow (BBF) planter or Raised bed method facilitates the crop to withstand heavy rains and moisture stress conditions and to obtain higher yields. Treat the seeds with carbendazim @ 1g or thiram or captan @ 3 g/kg followed by 1.5 ml of imidacloprid (48 %) and Rhizobium culture @ 200 g/10 kg seed. Apply 50 kg DAP + 26 kg MOP at the time of sowing and top dressing of urea @ 25 -30 kg/acre at 30 DAS. Pre-emergence herbicides like pendimethalin 30 % @ 1 - 1.5 L/ acre should be applied within 24-48 hours after sowing with the help of hand sprayer for better control of weeds. Post emergence application of quizalofop ethyl @ 400 ml/ acre for control of grasses and imazethapyr + imazomax 40 g /acre or imazethapyr 10 % @ 250 ml/acre to control the grasses and broad-leaved weeds. To control stem fly and stem gridler as a prophylactic application of carbofuran 3G granules @ 8 kg/acre is recommended at 10 DAS. To control of pod blight spraying of tebuconazole + sulphur @ 2.5 g/L is recommended. Dry the seed and maintain the below 12 % moisture during seed storage to avoid incidence of pest and disease.

Castor

Sow high yielding castor hybrids viz., ICH-5, ICH-66, PCH-111, DCH-519 & ICH-6, PCS-262 from 15th June to 15th July in rainy season under rainfed conditions. Intercrop with redgram in 1:1 or greengram in 1:3 ratio, open furrows between rows to conserve moisture in light soils.

LIVESTOCK

Animal Husbandry & Fisheries

Agriculture and allied sectors play a vital role in the lives and livelihoods of the majority of Telangana's population. The state is rich in livestock resources and

livestock production, according to national Livestock census 2019, the total livestock population in the state is 3.26 crore. The livestock sector is a major driver of primary sector growth. The sectors contribution to Telangana's Gross State Value Added (GSVA) at current prices increased from Rs. 96,908 crores in 2023-24 to Rs. 1,02,835 in 2024-25. Across India, inland fisheries where farmers build ponds on their landholdings to breed fish, is emerging as an important source of additional income to the farmers (Socio Economic Outlook, 2025). Scarcity of feeds and fodder resources is a big problem limiting animal production. Among all, flood and drought are the major devastating natural calamity leading to a massive loss of vegetation.

General Guidelines for Livestock Management: To protect animals from the weather fluctuations keep them in well-ventilated shelters and under the shade of trees. Restrict the movement of visitors to the livestock farms and practice biosecurity measures in case of disease out breaks. Keep sufficient feed, fodder and medicine in hand. Vaccinate animals against Foot and Mouth Disease and Hemorrhagic septicemia (Cows and buffaloes), PPR (goats). Except pregnant animals, deworm all animals for external and internal parasites. Keep clean drinking water available at all times. Monitor water intake by animals to protect from heat stress. Supplement herbal immune boosting products in ration to boost immunity. Observe and detect oestrus and inseminate on time. Splash/bath buffaloes especially pregnant animals to prevent adverse effect of summer on oestrus and conception and prevent early embryonic mortality/ abortion due to heat waves.

Offer ample milk or electrolyte water to new born calves to prevent dehydration, heat stroke and death of neonatal stocks. Maintain clean, well-ventilated shelters with proper drainage to prevent diseases during floods. Ensure adequate space: Overcrowding causes stress and disease outbreaks. Local authorities may be informed about the outbreak of the diseases. All animals should be untied/evacuated immediately in case of floods to avoid mortality. All dead animals should be disposed of hygienically and should not be thrown in nearby drainages, water bodies etc. Burial Pit should be of 8X9 feet. Sprinkling of disinfectants like bleaching powder, quick lime etc. in the affected areas helps in reducing the spread of diseases. Remove all the flammable items from the area in and around the animal shed. Always maintain a sufficient stock of feed, fodder, water and medicines and store them in a safe place. Proper Care should be taken after reallocation of animals after natural disasters. During Drought bajra fodder varieties and in water logging areas Para grass may be cultivated. Clean, safe, hygienic water should be provided to the animals. Restrict the transportation of animals to consumer's doorstep for milking.

Dairy

Disinfect sheds regularly with Phenol 1-2% solution and Washing Soda (4% solution). Deworm all adult stock with broad spectrum antihelmintic, Albendazole (Dose: 10 mg/ kg body weight) during last week of May. Vaccinate animals with Haemorrhagic septicaemia (HS) and black quarter in June @ 5ml per animal subcutaneously. Segregate animals in milk production, dry, pregnant, non- pregnant and heifers. Allow animals to graze early in morning or later in evening. Follow body weight-based feeding regimen. Provide 1 kg concentrate for every 2.5 kg milk for

cows and for every 2 kg milk for buffaloes. Feed colostrum to new born calf within twelve hours calving @ 1/10th of body weight. Provide tender leafy fodder to the calves for early development of rumen. Feed calf starter from 3rd day onwards to 2-3 weeks before weaning. For calves up to 3-6 months of age, mineralised salt blocks may be hung in the sheds to prevent mineral deficiency.

Deworm on 14th, 35th and 56th day followed by monthly up to six months and half yearly thereafter. Supplement with Area specific mineral mixture developed by PVNRTGVU @ 50-100g/day/animal and salt licks to prevent deficiencies. Avoid feeding spoiled or mouldy feed. Supplementation of Sodium Bi-carbonate (Baking soda) @ 30-50 g/day/animal for better milk fat yield and to avoid SARA. Azolla supplementation may be taken-up @1-2 Kg/dairy cattle per day. Do not allow animals for grazing at a time between 9am to 5pm (particularly period during heat wave period). During feed scarcity period feeding strategies such as densified Complete feed block@ 6-8 Kg/ day of these pellets sustain body maintenance and 3-4 kg of milk/day, Urea molasses mineral blocks, Liquid urea molasses feeding, feeding of urea treated straws, use of dry and fallen tree leaves like subabul, Mahua, banyanan, neem, mango (@ 50 kg of tree leaves 5 kg of GNC, 25kg of babul pods, 15 kg of molasses, 1kg of urea and 2 kg of mineral mixture) conventional and unconventional feeds can be offered to balance the feeding with locally available resources. The dry fodder like paddy straw and sorghum stover may be enriched with urea or molasses and salt to enhance the digestibility in large ruminants. Hydroponics fodder may be utilized wherever available. Isolate the sick animals and quarantine the newly purchased animals.

Small Ruminants (Sheep and Goat)

Animals should be kept under the shade of trees during the hot periods of the day. The size of the flock shall be reduced by culling Unproductive and un healthy animals should be sold. Sick animals should be sold as sick animal in the flock is a dead animal. Rotational deworming at quarterly interval. In pre-monsoon period, deworm the whole flock with broad spectrum anti-helminthic Albendazole @ 10mg/ kg body weight. Before the onset of monsoon, based on worm load by dung examination, deworming to be done with appropriate deworming medicine. Vaccinate with ET @ 2ml subcutaneously during May to June months in sheep. Blue tongue vaccination @ 1 ml intramuscularly should be done between July and August. FMD vaccine during the month of July/ August in sheep and goats. Smoke with neem leaves in the night time to control mosquito borne diseases. Always Keep the sheds clean and dry. Hang mineralized salt lick @1 block for 5 number of animals. Provide ad libidum water. Azolla supplementation may be taken-up @ 250-500 g/sheep/goat per day. Dipping with acaricides should be carried out to get rid of ectoparasites (ticks/fleas) in small ruminants. Fodder tree seedlings should be cultivated by using the rainfall and green fodder leaves may be fed to the animals. 250-300 gms of concentrate feed should be given to the pregnant ewes to avoid stillbirth or weak kids which is common in June, July months due to scarcity of pasture in grazing.

Poultry

Protect birds from excess summer heat. Vaccinate for Fowl Pox @ 0.5 ml intramuscularly and Ranikhet disease @ 0.5 ml subcutaneously. Add anti-stress

liquids & probiotics in drinking water during summer. Utilize broken rice or bajra (up to 30-40%), cotton seed meal (10-12%) and rapeseed meal (5 -8%) in poultry when there is shortage of maize during natural calamities. Keeping in view of disease outbreak follow biosecurity measures and mass vaccinate backyard poultry birds against Newcastle and viral diseases to prevent outbreak during stress periods. For hatching purpose of fertile eggs keep it in fridge at temperature of 4°C. To improve the growth rate 1 per cent protein level (soybean meal) may be added in the feed. Suitable coccidiostat should be added in the feed continuously by consulting the local veterinarian to prevent coccidiosis. 3-5 g of oyster shell/limestone/grit per day/bird shall be given to laying hens to avoid leathery eggs.

Value Addition: Convert milk into ghee, paneer, curd, khoa for better returns. Explore meat processing and packaging for hygienic sale. Utilize manure for biogas and compost.

Fisheries

Failed Monsoon: Use bore well or alternative water sources to maintain minimum pond water levels. Reduce stocking density to lower water demand. Adopt partial harvest strategies to minimize losses. Use water-saving practices like reduced water exchange. Avoid new stocking if water availability is uncertain.

Flood: Strengthen pond bunds and spillways to prevent overflow and escape of fish. Install protective netting around pond perimeters. Avoid feeding during peak flood to prevent water quality issues. Harvest fish early if flood risk is imminent. Apply disinfectants like Benzalkonium Chloride 80% (BKC) @ 2.5 L/ha and Rock Salt @ 75-125 kg/ha/5ft depth/ha after floodwater recedes to control disease outbreaks.

Drought: Reduce biomass load in ponds by early or partial harvesting. Avoid use of organic manure if water level is too low to prevent oxygen depletion. Apply aeration measures (e.g., paddle wheel aerators) to maintain DO levels. Maintain good water quality by regular monitoring and minimum exchange.

Delayed Monsoon: Postpone seed stocking until adequate water level is available. Increase application of manures/fertilizers once rains begin to promote plankton growth. Avoid breeding activities during uncertainty to protect brood stock. Maintain brood stock in protected tanks.

Advanced Monsoon: Prepare ponds early for stocking, complete liming 200 kg/ha and fertilization in advance. Ensure brood stock is not disturbed during monsoon (15th June to 31st August). Avoid overfeeding as heavy rains can cause feed waste and water quality degradation. Watch for sudden temperature or pH fluctuations due to early rains.

High Temperature: Maintain higher water levels (5-6 feet) to buffer temperature fluctuations. Avoid feeding during peak heat hours, feed early morning and late evening. Use shade nets or plant trees around ponds to reduce direct sunlight. Monitor dissolved oxygen (DO) and apply aeration during hot afternoons.

Low Temperature: Reduce feeding rate and switch to highly digestible feed. Avoid handling fish during early mornings and nights. Provide windbreaks and increase pond depth to retain heat. Avoid use of antibiotics and strong chemicals during cold periods unless essential.